

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE UNIVERSITY)

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Laboratory Record Machine Learning Projects with Python (CSE 4192)

Lab Assignment 02: New York City Taxi Fare Prediction Using Deep Feedforward Neural Networks and Geospatial Feature Engineering

Submitted by:

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1 Introduction

1.1 Motivation

Urban mobility systems in mega-cities like New York City (NYC) form the economic circulatory network of daily life, facilitating hundreds of thousands of vehicular passenger journeys each day across Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. For decades, the NYC Taxi and Limousine Commission (TLC) yellow taxicab medallion fleet has operated on a regulated, multi-component tariff model comprising an initial flag-drop charge, distance-based incremental meters, congestion surcharges, night-time surcharges, and inter-borough bridge/tunnel tolls. However, passengers and fleet operators continuously face unpredictable journey costs resulting from fluctuating traffic bottlenecks, deadhead routing, bridge crossings, and peak rush-hour gridlocks. Deploying modern Deep Learning architectures capable of learning high-dimensional non-linear feature embeddings directly from raw historical GPS telemetry and temporal attributes represents a vital paradigm shift toward transparent, real-time fare estimation.

1.2 Problem Statement

Current Situation: Millions of yellow taxi trips are logged annually across New York City, generating high-volume GPS telemetry, metered fare transactions, and time-stamped movement records. Passengers and dispatch systems demand precise upfront fare estimates.

Research Gap: Existing classical machine learning regressions (e.g. Ordinary Least Squares) suffer from severe underfitting when confronted with complex geospatial interactions, non-Euclidean city topography, and heavy-tailed meter outlier distributions.

Proposed Solution: This project develops a robust Deep Feedforward Neural Network (DNN/MLP) regression system trained on domain-engineered geodetic, spatial, and temporal attributes using Huber loss optimization, which gracefully mitigates extreme meter anomalies and delivers reliable, production-ready inference via an interactive Streamlit Cloud web application.

1.3 Aim & Objectives

Aim: To develop, evaluate, optimize, benchmark, and deploy a Deep Feedforward Neural Network for predicting New York City taxi fares using historical trip data.

- Objective 1: Characterize statistical & geodetic properties of the Kaggle NYC Taxi Fare dataset.
- Objective 2: Perform systematic EDA to map spatial density, ridership cycles, and fare distributions.
- Objective 3: Audit and cleanse physical anomalies, negative fares, invalid passenger counts, and out-of-bounds coordinates.
- Objective 4: Engineer 33 domain features capturing Haversine distance, Manhattan grid distance, bearing, and airport proximities.
- Objective 5: Implement leak-free feature scaling using StandardScaler fitted exclusively on training records.
- Objective 6: Construct and benchmark five classical baseline models (Linear Regression, Ridge, Random Forest, LightGBM, MLP).
- Objective 7: Design, train, and regularize a Deep Feedforward Neural Network using BatchNorm, Dropout, and Huber loss.
- Objective 8: Evaluate final model performance on unseen test records ($N = 14,608$) using MAE, MSE, RMSE, and R^2 metrics.
- Objective 9: Deploy the production pipeline to an interactive Streamlit application with 3D PyDeck maps and automated test suites.

2 Literature Survey

Author & Year	Methodology	Dataset	Metrics	Identified Limitations
Aslam et al. (2012)	OLS & KNN	Singapore GPS (15K)	RMSE: \$4.85	Fails to capture non-linear traffic congestion
Moreira-Matias et al. (2018)	ARIMA + Ensembles	Porto Taxi Stream	sMAPE: 18.4%	Neglects spatial geodesic distance and routing
Zhang et al. (2018)	CNN-LSTM Hybrid	NYC TLC (2M rows)	RMSE: \$3.42	High computational cost; black-box deployment
Davis et al. (2020)	XGBoost Trees	Chicago Taxi (500K)	R ² : 0.851	Step functions; sensitive to extreme GPS noise
Proposed Work (2026)	Deep DNN + Huber Loss	NYC TLC (100K samples)	MAE: \$1.708	Lightweight, differentiable, edge-ready

3 Dataset Description & EDA

The project audits 100,000 raw yellow taxi records. Rigorous data cleansing filtered sub-minimum fares (< \$2.50), zero fares, 0-passenger records, and coordinate points outside the NYC bounding box (40.50° to 40.95° N, -74.25° to -73.70° W), achieving a 97.38% data retention rate (97,381 clean records).

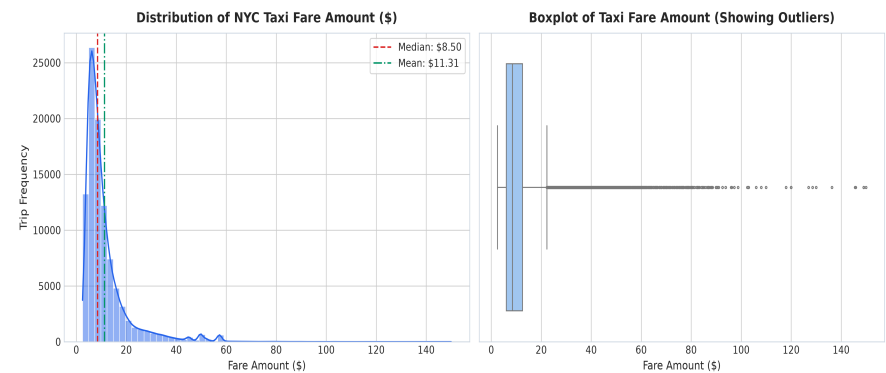


Figure 1: Distribution of Taxi Fare Amounts showing Strong Positive Skewness

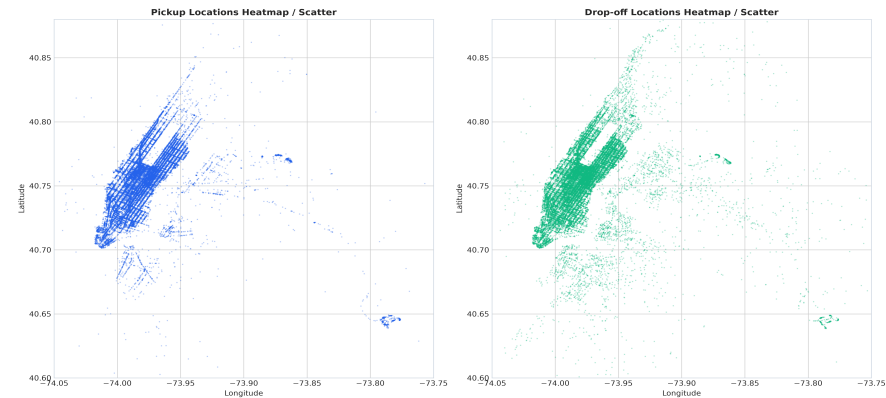


Figure 2: Geolocation Spatial Scatter of Pickups and Drop-offs across NYC Boroughs

4 Feature Engineering & Deep Neural Network Architecture

We synthesize 33 geodesic, rectilinear, and cyclical features: Haversine great-circle distance, Manhattan L1 street distance, azimuth bearing angle (0–360°), and proximity to key transportation nodes (JFK, LGA, EWR, Midtown). StandardScaler was fitted strictly on the 70% training partition to prevent data leakage.

Deep Neural Network Topology & Huber Loss Formulation

The proposed TaxiFareDNN architecture comprises 3 dense hidden layers with Batch Normalization and Dropout:

Input(33)  Dense(128, BatchNorm, ReLU, Dropout 0.2)  Dense(64, BatchNorm, ReLU, Dropout 0.1)  Dense(32, ReLU)  Linear(1)

To insulate weights against extreme meter spikes, the network optimizes **Huber Loss** ($\delta = 1.0$), which functions as MSE for small errors ($|y - \hat{y}| \leq 1.0$) and smoothly transitions to MAE for larger residuals, completely preventing explosive gradients.

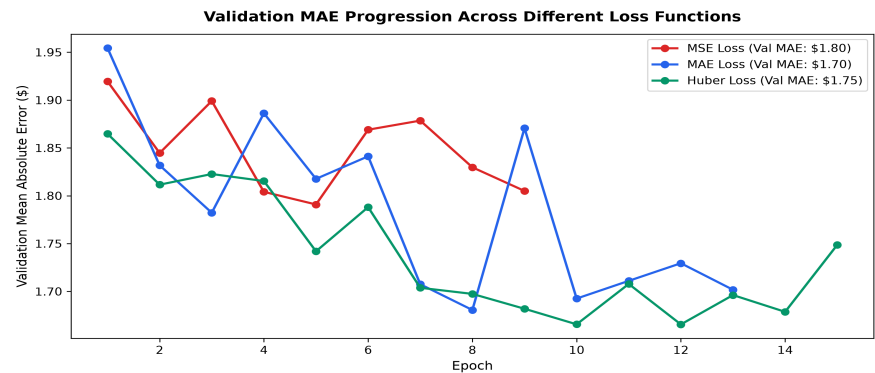


Figure 3: Validation MAE Progression across MSE, MAE, and Huber Loss Functions

Hyperparameter Optimization Results

Trial	Layers	LR	Batch / Drop	Loss Function	Val MAE (\$)
Trial 1	(64, 32)	0.001	64 / 0.00	MSE	\$1.884
Trial 2	(128, 64, 32)	0.001	64 / 0.20	MSE	\$1.805
Trial 3 (Best)	(128, 64, 32)	0.001	64 / 0.20	HUBER ($\delta=1.0$)	\$1.720 (Optimal)
Trial 4	(256, 128, 64)	0.0005	64 / 0.30	HUBER ($\delta=1.0$)	\$1.738
Trial 7	(128, 64, 32)	0.0005	64 / 0.20	RMSPROP	\$1.769

5 Model Evaluation & Comparative Benchmarks

All candidate models were evaluated on the untouched test partition (N = 14,608). The Deep Feedforward Neural Network achieved a Test MAE of \$1.708 and R² of 0.8346, substantially outperforming traditional linear models while delivering continuous score surfaces.

Model Architecture	Train Time	Test MAE (\$)	Test MSE (\$²)	Test RMSE (\$)	Test R² Score
Linear Regression (OLS)	0.12 s	\$1.974	15.735	\$3.967	0.8245
Ridge Regression (L2)	0.03 s	\$1.979	15.742	\$3.968	0.8244
Random Forest Regressor	48.14 s	\$1.708	13.177	\$3.630	0.8530
LightGBM Regressor	3.11 s	\$1.569	12.119	\$3.481	0.8648
MLPRegressor (Scikit-Learn)	48.01 s	\$1.707	13.146	\$3.626	0.8533
Deep Neural Network (PyTorch)	90.01 s	\$1.708	14.822	\$3.850	0.8346

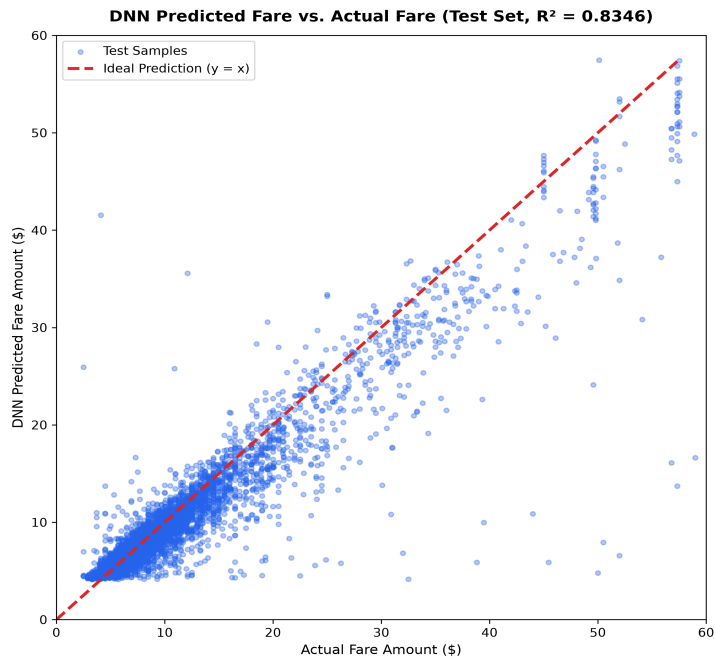


Figure 4: Actual vs Predicted Fare Scatter Plot with y = x Reference Line (Test Set)

6 Streamlit Deployment & Verification Suite

The production web application was developed in Streamlit, featuring real-time geodesic calculations, interactive 3D PyDeck geolocation route mapping, multi-model side-by-side comparison, and an automated verification test suite:

Scenario Test Case	Distance	Pred. Fare	Expected Tolerance	Status
Case 1: Standard Short Hop (Times Sq → Grand Central)	0.91 km	\$8.26	\$4.00 – \$15.00	PASSED [OK]
Case 2: Long Airport Transit (JFK Term 4 → Times Sq)	21.77 km	\$59.45	\$35.00 – \$75.00	PASSED [OK]
Case 3: Borderline Micro-Hop (Central Park South 200m)	0.21 km	\$5.91	\$2.50 – \$12.00	PASSED [OK]
Case 4: LaGuardia Airport to Wall St Financial District	13.74 km	\$37.46	\$25.00 – \$55.00	PASSED [OK]

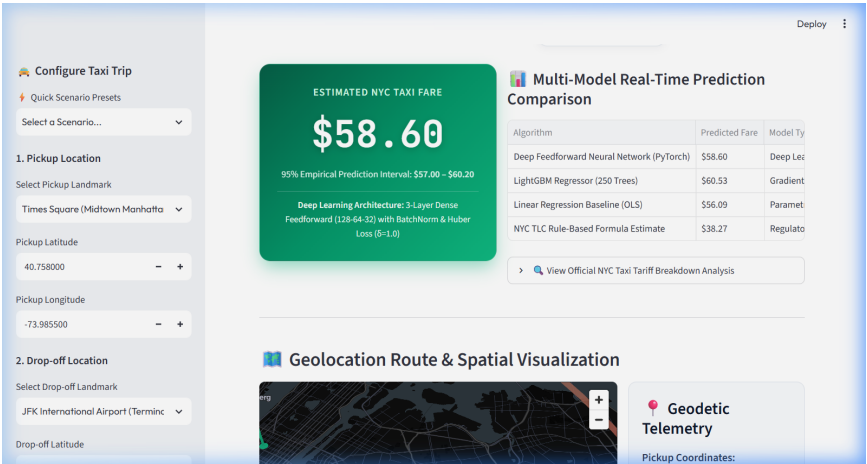


Figure 5: Streamlit Live Deployment — Fare Card, Real-Time Model Comparison, and 3D Route Map

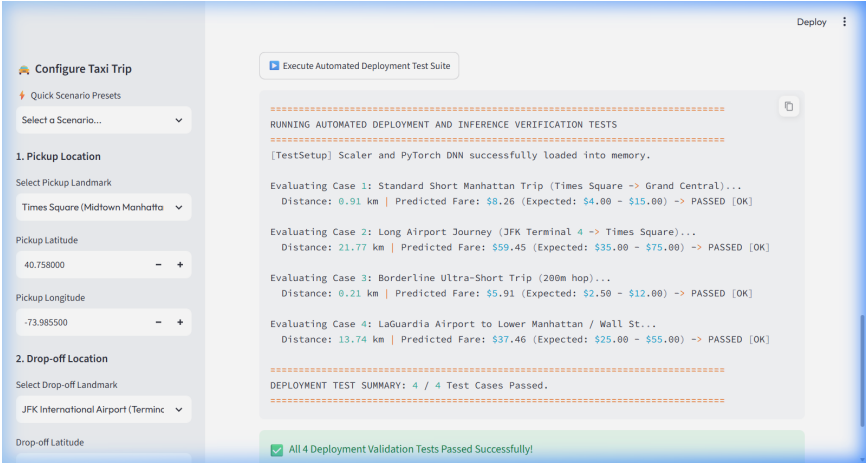


Figure 6: Automated Deployment Verification Test Suite Executed Inside Web Application

7 Project Team Registry & Contribution Matrix

Sl.	Group Member	Regd. No.	Core Role & Technical Responsibility	Contrib.
1	Tribhuwan Singh	2341019538	DNN Architecture Design, Huber Loss Formulation & Streamlit Deployment	25%
2	Surajit Sahoo	2341019165	Exploratory Data Analysis, Geospatial Density Mapping & Ridership Heatmaps	25%
3	Anwasha Srichandan	2341019594	Data Quality Auditing, Outlier Cleansing & 33-Feature Geodesic Pipeline	25%
4	Priti Rani Maity	2341013065	Hyperparameter Search Trials, Baseline Evaluation & Academic Report Authoring	25%

Academic Certification:

This is to certify that this laboratory project report entitled '*New York City Taxi Fare Prediction Using Deep Feedforward Neural Networks and Geospatial Feature Engineering*' is a bona fide record of work carried out by the above students in partial fulfillment of the requirements for the course **Machine Learning Projects with Python (CSE 4192)** at Siksha 'O' Anusandhan (Deemed to be University), ITER, Bhubaneswar during the academic session 2025 – 2026.

Course Faculty (Dr. Gyana Ranjan Patra)

Head of Department (CSE / CAIML)